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Original Article

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# Pharmacokinetic comparison between the fixed-dose combination of rosuvastatin/ezetimibe 2.5/10 mg and the co-administration of individual formulations in healthy subjects

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**Trial Registration**
Clinical Research Information Service

## ABSTRACT

The combination therapy of low-dose rosuvastatin and ezetimibe has shown similar efficacy in decreasing low-density lipoprotein cholesterol (LDL-C) compared to high-dose rosuvastatin monotherapy. This study aimed to compare the pharmacokinetics (PKs) between the fixed-dose combination (FDC) of rosuvastatin/ezetimibe 2.5/10 mg and the co-administration of individual formulations. A randomized, open-label, single-dose, 2-sequence, 2-period, crossover study was conducted in healthy volunteers. Subjects were randomized in each sequence and received a single dose of the FDC of rosuvastatin/ezetimibe 2.5/10 mg or the co-administration of individual formulations in each period with a 14-day washout. Serial blood samples for PK analysis were collected up to 48 hours post-dose for rosuvastatin and 72 hours post-dose for ezetimibe. The geometric mean ratios (GMRs) and 90% confidence intervals (CIs) of the FDC to the co-administration for maximum plasma concentration ($C_{max}$) and area under the curve from zero to the last measurable time point ($AUC_{last}$) were calculated. Forty-seven subjects were randomized, and 41 subjects completed the study. The GMRs (90% CIs) of $C_{max}$ and $AUC_{last}$ for rosuvastatin were 0.9789 (0.9020–1.0622) and 0.9741 (0.9105–1.0422). The corresponding values were 1.0419 (0.9219–1.1776) and 0.9983 (0.9522–1.0465) for total ezetimibe, and 1.0396 (0.9087–1.1893) and 0.9743 (0.8997–1.0550) for free ezetimibe, respectively. All the values were within the conventional bioequivalence criteria of 0.8 to 1.25. In conclusion, the FDC of rosuvastatin/ezetimibe 2.5/10 mg showed comparable PK with the co-administration of individual formulations.

**Trial Registration**: Clinical Research Information Service Identifier: KCT0009254

**Keywords:** Rosuvastatin; Ezetimibe; Drug Combinations; Pharmacokinetics

Identifier: KCT0009254

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**Conflict of Interest**

- Authors: Dowon Lee, Sejung Hwang, Kyung-Sang Yu, and SeungHwan Lee declare no conflicts of interest related to this study and Taewon Lee is currently employed by the Addpharma Co., Ltd. (Seongnam, Korea).
- Reviewers: Nothing to declare
- Editors: Nothing to declare

**Author Contributions**

Conceptualization: Lee S; Supervision: Hwang S, Lee T, Yu KS, Lee S; Visualization: Lee D; Writing - original draft: Lee D.

# INTRODUCTION

Dyslipidemia, characterized by abnormal serum levels of cholesterol, triglycerides, or both, involving abnormal levels of related lipoprotein species, is diagnosed in 40% of the Korean population [1,2]. The most commonly associated clinical consequence of dyslipidemia is increased atherosclerotic cardiovascular disease (ASCVD) risk [1]. Statin is the first-line therapy for reducing the risk of ASCVD by lowering low-density lipoprotein cholesterol (LDL-C) levels in the blood through the inhibition of 3-hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase, through directly reducing intracellular cholesterol biosynthesis [3,4]. Rosuvastatin is the synthetic statin approved by the U.S. Food and Drug Administration (FDA) in 2003 for the treatment of hypercholesterolemia and mixed dyslipidemia with a recommended dosage of 5–40 mg once daily [5,6]. In Asian populations, where systemic exposure to rosuvastatin is higher than Caucasians, a lower dose of rosuvastatin 2.5 mg has been approved and is recommended as an initial dose to mitigate the risk of statin-associated adverse reactions such as myopathy [5,7,8].

Ezetimibe, one of the lipid-lowering agents, works alone or synergistically with statins by inhibiting intestinal cholesterol absorption by selectively blocking the Niemann-Pick C1-like 1 (NPC1L1) protein [9-11]. Ezetimibe is extensively metabolized into its active glucuronide form, which accounts for approximately 80–90% of the total drug in plasma [12]. Both the parent drug and its glucuronide metabolite exhibit comparable pharmacological activity [13]. The glucuronide metabolite undergoes enterohepatic circulation, contributing to prolonged pharmacological effect and enabling once-daily dosing [14].

Combination therapy with statins and other lipid lowering agents has been widely used to improve the achievement of LDL-C goals with different mechanisms addressing the limitations of statin monotherapy [15]. Among these combinations, therapy with statin and ezetimibe also enhances other lipid levels, including total cholesterol, non-high-density lipoprotein cholesterol, and apolipoprotein B, with no significant pharmacokinetic (PK) drug interaction observed [11,16]. One study conducted to evaluate the efficacy of the combination therapy of 2.5 mg rosuvastatin and 10 mg ezetimibe compared to 5 mg rosuvastatin monotherapy in patients with type 2 diabetes mellitus and hypercholesterolemia revealed that add-on therapy with ezetimibe achieved not only quantitative but also qualitative improvement of serum lipid profile [17].

Adherence to lipid-lowering therapy is important for patients with cardiovascular disease because poor compliance has been associated with worse clinical outcomes and increased cardiovascular morbidity and mortality [18]. Fixed-dose combination (FDC) has been shown to be an effective method for reducing the risk of medication non-adherence by 26% compared to free-drug component regimens, thereby improving medication compliance [19]. While FDCs combining ezetimibe 10 mg with higher doses of rosuvastatin (5–20 mg) have already been developed and approved, the 2.5 mg dose of rosuvastatin is used in Asian populations, particularly among patients who are statin-intolerant or require only a mild reduction in LDL-C levels [9]. Given this clinical context, the development of an FDC containing rosuvastatin 2.5 mg and ezetimibe 10 mg was warranted to provide a low-dose therapeutic option that enhances adherence and meets the needs of specific patient populations.

Therefore, the FDC of rosuvastatin/ezetimibe 2.5/10 mg was developed to effectively lower LDL-C levels while improving patient compliance and ensuring efficacy and safety. This study


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aimed to evaluate the comparative PKs of the FDC of rosuvastatin/ezetimibe 2.5/10 mg and the co-administration of rosuvastatin 2.5 mg and ezetimibe 10 mg in healthy Korean volunteers.

# METHODS

## Ethics

This study was approved by Korean Ministry of Food and Drug Safety (MFDS) and the Institutional Review Board of H Plus Yangji Hospital (Seoul, Korea). In addition, the study was conducted in accordance with ethical principles of the Declaration of Helsinki, and International Conference on Harmonization Guidelines for Korean Good Clinical Practice. This study was registered in the Clinical Research Information Service (KCT0009254). Only volunteers who signed written informed consent were included in the study.

## Study population

Healthy adults aged more than 19 years who had a total body weight ≥ 50 kg for men or ≥ 45 kg for women, with a body mass index (BMI) between 18.0 and 30.0 kg/m$^2$ were eligible to participate in this study. Subjects were screened within 30 days prior to the first administration of investigational product (IP) to assess whether they met the inclusion and exclusion criteria, which involved history taking including drinking, caffeine, and smoking habits as well as physical examination, clinical laboratory tests, vital signs, and 12-lead electrocardiogram (ECG).

## Study design

This was a randomized, open-label, single-dose, 2-sequence, 2-period, crossover study. Eligible subjects were randomized into sequence A or B in the first period. Sequence A was the co-administration of individual formulations followed by the FDC, and sequence B was the FDC followed by the co-administration. In the first period, all subjects received a single dose of the FDC of rosuvastatin/ezetimibe 2.5/10 mg or the individual formulations of rosuvastatin 2.5 mg and ezetimibe 10 mg with 150 mL of water on the fasting state according to the sequence. After a 14-day washout period, subjects received an alternative treatment in the same manner with the first period.

## Sample size

The intra-individual coefficient of variation (CV) of maximum plasma concentration ($C_{max}$) and area under the curve from zero to the last measurable time point ($AUC_{last}$) for total ezetimibe was calculated to be 20.43% and 13.25%, and the intra-individual CV of $C_{max}$ and $AUC_{last}$ for rosuvastatin was calculated to be 19.46% and 14.91% from literature [20]. Assuming the highest intra-individual CV of 21% and based on the FDC to co-administration ratio of 0.90 with 80% statistical power at a 5% significance level, 40 subjects were required to show whether the PK parameters were different by more than 20% between treatments. Considering a dropout rate of 20%, the target number of subjects was determined to be 50.

## PK assessments

Blood samples for plasma PK parameters were collected at 0 (pre-dose), 0.25, 0.5, 0.75, 1, 1.25, 2, 3, 4, 5, 6, 8, 12, 24, and 48 hours post-dose for rosuvastatin and at 0 (pre-dose), 0.25, 0.5, 0.75, 1, 2, 3, 4, 6, 8, 12, 24, 48, and 72 hours post-dose for total ezetimibe and free ezetimibe in each period. Approximately 4 mL and 6 mL of blood were collected at the time points allocated to rosuvastatin and ezetimibe measurements, respectively, and


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a total of 10 mL at the time points when both were measured. The collected samples were immediately placed in an ice bath, and centrifuged at 4°C, 3000 rpm for 10 minutes. Plasma was aliquoted into polypropylene tubes and stored at −70°C until analysis. The analysis of plasma concentrations of rosuvastatin, total ezetimibe and free ezetimibe was performed by a validated ultra-performance liquid chromatography-tandem mass spectrometry method in accordance with the European Medicines Agency Guideline on Bioanalytical Method Validation (EMEA/CHMP/EWP/192217/2009 Rev.1) [21].

PK parameters were calculated by non-compartmental methods using Phoenix WinNonlin software (Version 8.0; CERTARA USA Inc., Radnor, PA, USA). $C_{max}$ and time of maximum plasma concentration ($T_{max}$) were directly obtained from observed data. $AUC_{last}$ was calculated using the linear trapezoidal model. Area under the curve from zero to infinity ($AUC_{inf}$) was calculated as $AUC_{last} + C_{last}/\lambda_z$, $C_{last}$ is the last plasma concentration measurement, and $\lambda_z$ is the elimination rate constant. Elimination terminal half-life ($t_{1/2}$) was calculated as $\ln(2)/\lambda_z$. Apparent total clearance after oral administration (CL/F) and apparent volume of distribution after oral administration ($V_d$/F) were calculated as dose/$AUC_{inf}$ and dose/$\lambda_z \cdot AUC_{inf}$, respectively.

### Safety and tolerability assessments

Safety and tolerability were assessed in all subjects who received IP at least once. Physical examinations, concomitant medications, clinical laboratory tests (hematology, blood chemistry, urinalysis), vital signs, 12-lead ECGs, and adverse events (AEs) monitoring were performed throughout the study. All the AEs, coded according to the Medical Dictionary for Regulatory Activities (MedDRA, version 24.1; International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use [ICH], Geneva, Switzerland), were summarized by treatment for seriousness, severity, relationships, result and action taken. The number of subjects, proportion (%) and count were also summarized by treatment for each of these categories.

### Statistical analysis

The statistical analysis was performed using SAS version 9.4 software (SAS Institute Inc., Cary, NC, USA). A generalized linear model was used for the comparison of PK parameters across the two different treatments, with log-transformed data. Period, sequence and treatment were set as fixed effects, while subjects nested within sequence were considered as a random effect. Through the model, the geometric least squares mean ratios (GMRs) and 90% confidence intervals (CIs) for $C_{max}$ and $AUC_{last}$ were calculated and evaluated if these values were within the conventional bioequivalence criteria of 0.8 to 1.25.

## RESULTS

### Study population

A total of 71 participants were screened, resulting in 47 participants being enrolled and randomized into each sequence (Sequence A: 22, Sequence B: 25). Forty-one subjects completed the study since 3 participants withdrew their consent and 3 participants were dropped out because of AEs during the study.

The enrolled 47 subjects consisted of 39 males and 8 females, with a mean age ± standard deviation of 29.38 ± 9.13 years. The body weight and BMI of the enrolled subjects were


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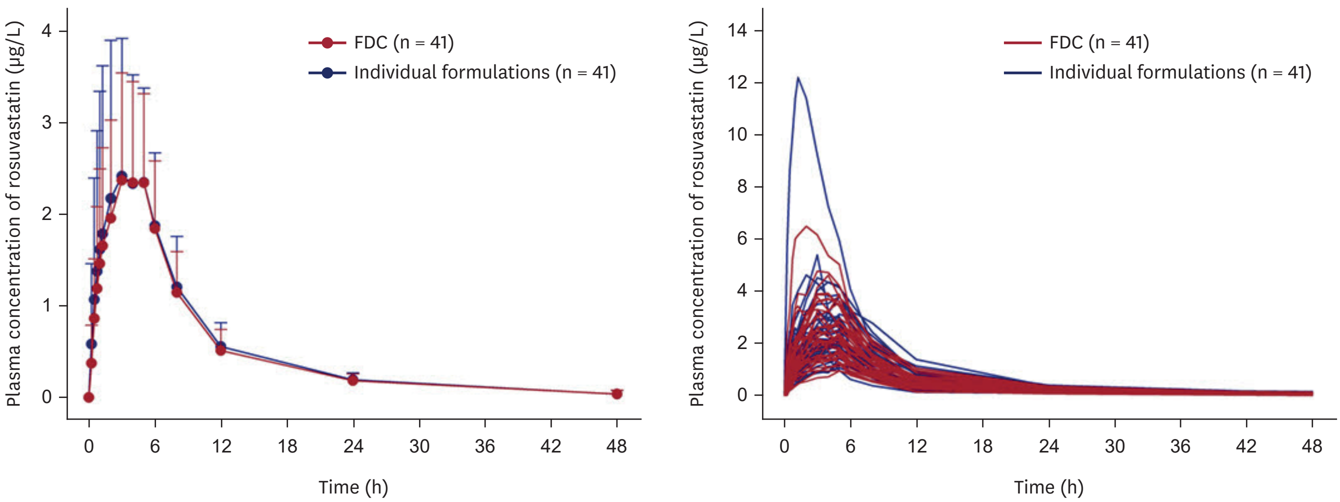


**Figure 1.** Plasma concentration-time profiles of rosuvastatin after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5/10 mg or the co-administration of individual formulations. Bars represent standard deviations (left panel: mean profiles, right panel: individual profiles).
FDC, fixed-dose combination.

70.26 ± 10.40 kg and 23.76 ± 2.76 kg/m², respectively. There were no statistically significant differences in demographic characteristics between the sequences.

### PK results

The mean concentration-time profiles of rosuvastatin were similar between the FDC and the co-administration of individual formulations (**Fig. 1**). Individual $C_{max}$ and $AUC_{last}$ of rosuvastatin after administration of the FDC and the co-administration of each formulation showed similarity (**Fig. 2**). The GMRs (90% CIs) for FDC to the co-administration of $C_{max}$ and $AUC_{last}$ for rosuvastatin were 0.9789 (0.9020–1.0622) and 0.9741 (0.9105–1.0422), respectively, and met the bioequivalence criteria of 0.8 to 1.25 (**Table 1**).

The mean concentration-time profiles of total ezetimibe and free ezetimibe were similar between the FDC and the co-administration of individual formulations (**Fig. 3**). Multiple peaks were observed in the individual concentration-time profiles, which is known to be due

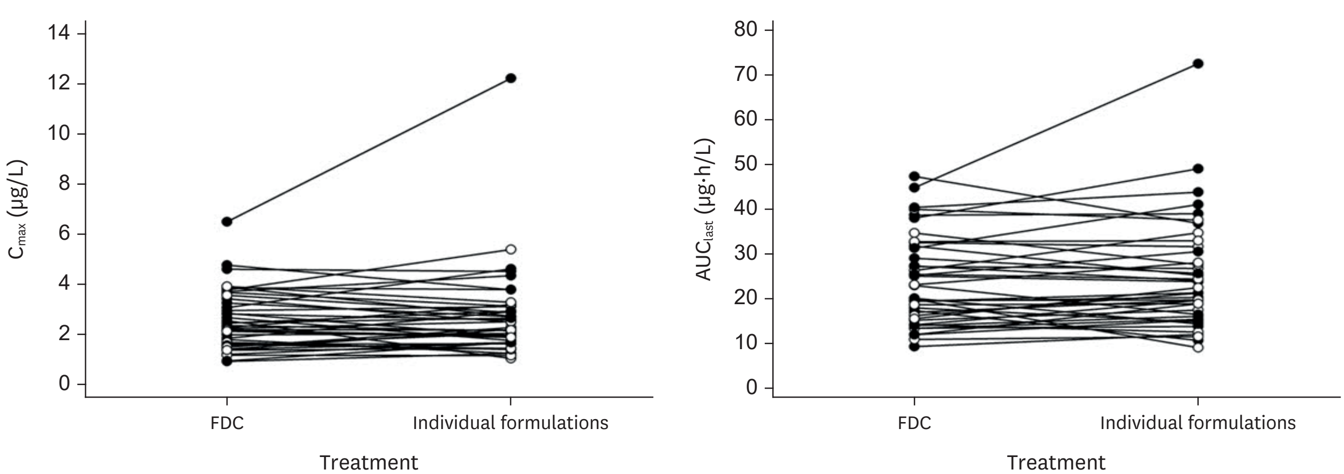


**Figure 2.** Individual comparison for $C_{max}$ and $AUC_{last}$ of rosuvastatin, after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations.
$C_{max}$, maximum plasma concentration; $AUC_{last}$, area under the curve from zero to the last measurable time point; FDC, fixed-dose combination.

Table 1. Summary of PK parameters of rosuvastatin after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations

| Parameter | FDC (n = 41) | Individual formulations (n = 41) | GMR* (90% CIs) |
|---|---|---|---|
| $C_{max}$ (μg/L) | 2.60 ± 1.19 | 2.74 ± 1.83 | 0.9789 (0.9020–1.0622) |
| $AUC_{last}$ (μg·h/L) | 23.72 ± 9.97 | 24.77 ± 12.33 | 0.9741 (0.9105–1.0422) |
| $AUC_{inf}$ (μg·h/L) | 25.01 ± 10.05 | 26.14 ± 12.36 | 0.9726 (0.9111–1.0382) |
| $T_{max}$ (h) | 4.00 (1.25–6.00) | 5.00 (0.75–6.00) | |
| $t_{1/2}$ (h) | 8.92 ± 4.66 | 8.52 ± 3.31 | |
| CL/F (L/h) | 117.00 ± 47.29 | 114.96 ± 48.96 | |
| $V_d$/F (L) | 1,463.77 ± 933.34 | 1,341.82 ± 644.83 | |

Values are represented as arithmetic mean ± standard deviation except for $T_{max}$, which is represented as arithmetic median (range).
PK, pharmacokinetic; FDC, fixed-dose combination; GMR, geometric mean ratio; CI, confidence interval; $C_{max}$, maximum plasma concentration; $AUC_{last}$, area under the curve from zero to the last measurable time point; $AUC_{inf}$, area under the curve from zero to infinity; $T_{max}$, time of maximum plasma concentration; $t_{1/2}$, terminal half-life; CL/F, apparent total clearance after oral administration; $V_d$/F apparent volume of distribution after oral administration.
*GMR is the geometric least squares mean ratio of the FDC to the co-administration of individual formulations.

to the enterohepatic circulation of ezetimibe (**Fig. 3**) [13]. Individual $C_{max}$ and $AUC_{last}$ of total ezetimibe and free ezetimibe after administration of the FDC and the co-administration of each formulation were similar (**Fig. 4**). The GMRs (90% CIs) for FDC to the co-administration of $C_{max}$ and $AUC_{last}$ 1.0419 (0.9219–1.1776) and 0.9983 (0.9522–1.0465) for total ezetimibe, and 1.0396 (0.9087–1.1893) and 0.9743 (0.8997–1.0550) for free ezetimibe (**Table 2**). The GMRs (90% CIs) for FDC to the co-administration of $C_{max}$ and $AUC_{last}$ for total ezetimibe and free ezetimibe were also within the conventional bioequivalence criteria of 0.8 to 1.25 (**Table 2**).

## Safety results

A total of 10 treatment emergent adverse events (TEAEs) occurred in 9 subjects (19.1%) (**Supplementary Table 1**). Eight TEAEs occurred in 7 subjects (15.2%) after the administration of the FDC and 2 TEAEs occurred in 2 subjects (4.7%) after administration

Table 2. Summary of PK parameters of total ezetimibe and free ezetimibe after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations

| Parameter | FDC (n = 41) | Individual formulations (n = 41) | GMR* (90% CIs) |
|---|---|---|---|
| Total ezetimibe | | | |
| $C_{max}$ (μg/L) | 86.09 ± 31.90 | 85.52 ± 36.94 | 1.0419 (0.9219–1.1776) |
| $AUC_{last}$ (μg·h/L) | 571.97 ± 256.26 | 578.18 ± 267.69 | 0.9983 (0.9522–1.0465) |
| $AUC_{inf}$ (μg·h/L) | 611.07 ± 263.32 | 640.66 ± 293.78 | 0.9668 (0.9151–1.0214) |
| $T_{max}$ (h) | 1.00 (0.50–4.00) | 1.00 (0.50–3.00) | |
| $t_{1/2}$ (h) | 17.04 ± 6.81 | 20.27 ± 14.20 | |
| CL/F (L/h) | 18.56 ± 6.00 | 18.25 ± 7.03 | |
| $V_d$/F (L) | 450.02 ± 232.99 | 494.56 ± 272.67 | |
| Free ezetimibe | | | |
| $C_{max}$ (μg/L) | 5.59 ± 3.50 | 5.52 ± 3.98 | 1.0396 (0.9087–1.1893) |
| $AUC_{last}$ (μg·h/L) | 80.41 ± 38.66 | 82.29 ± 39.61 | 0.9743 (0.8997–1.0550) |
| $AUC_{inf}$ (μg·h/L)† | 87.29 ± 42.23 | 94.33 ± 48.36 | 0.9345 (0.8493–1.0283) |
| $T_{max}$ (h) | 1.00 (0.25–8.00) | 1.00 (0.50–24.00) | |
| $t_{1/2}$ (h)† | 17.89 ± 7.64 | 21.53 ± 15.36 | |

Values are represented as arithmetic mean ± standard deviation except for $T_{max}$, which is represented as arithmetic median (range).
PK, pharmacokinetic; FDC, fixed-dose combination; GMR, geometric mean ratio; CI, confidence interval; $C_{max}$, maximum plasma concentration; $AUC_{last}$, area under the curve from zero to the last measurable time point; $AUC_{inf}$, area under the curve from zero to infinity; $T_{max}$, time of maximum plasma concentration; $t_{1/2}$, terminal half-life; CL/F, apparent total clearance after oral administration; $V_d$/F apparent volume of distribution after oral administration.
*GMR is the geometric least squares mean ratio of the FDC to the co-administration of individual formulations.
†n = 40, As the elimination rate constant of one subject could not be determined, one subject was excluded from the descriptive statistics for $AUC_{inf}$ and $t_{1/2}$.


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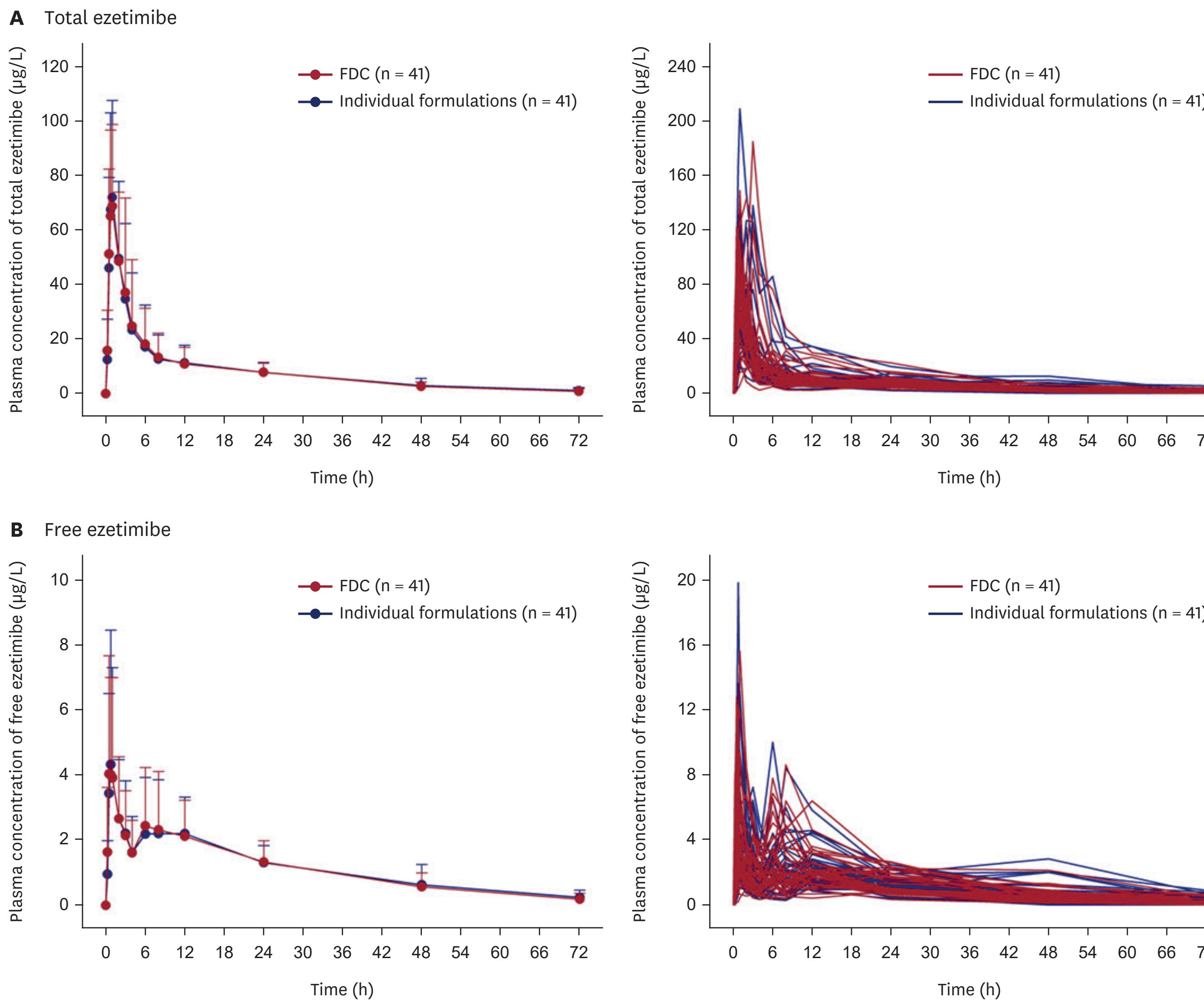


**Figure 3.** Plasma concentration-time profiles of (A) total ezetimibe and (B) free ezetimibe after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations. Bars represent standard deviations (left panel: mean profiles, right panel: individual profiles). FDC, fixed-dose combination.

of the co-administration of individual formulations (**Supplementary Table 1**). Following the administration of FDC, 1 serious adverse event (SAE), severe acute respiratory syndrome coronavirus 2 test positive, was reported in 1 subject (**Supplementary Table 1**). In addition, 3 subjects withdrew from this study due to 4 TEAEs: creatinine phosphokinase (CPK) increased (moderate), alanine transaminase increased (mild, n = 2), and gamma-glutamyl transferase increased (mild). All of these events resolved spontaneously without pharmacological intervention, except for the case of CPK increased, for which the outcome could not be confirmed due to follow-up loss.

All TEAEs were considered study drug related and evaluated as mild to moderate in severity, except for the SAE. No subject showed clinically significant changes in 12-lead ECGs and vital sign measurements.


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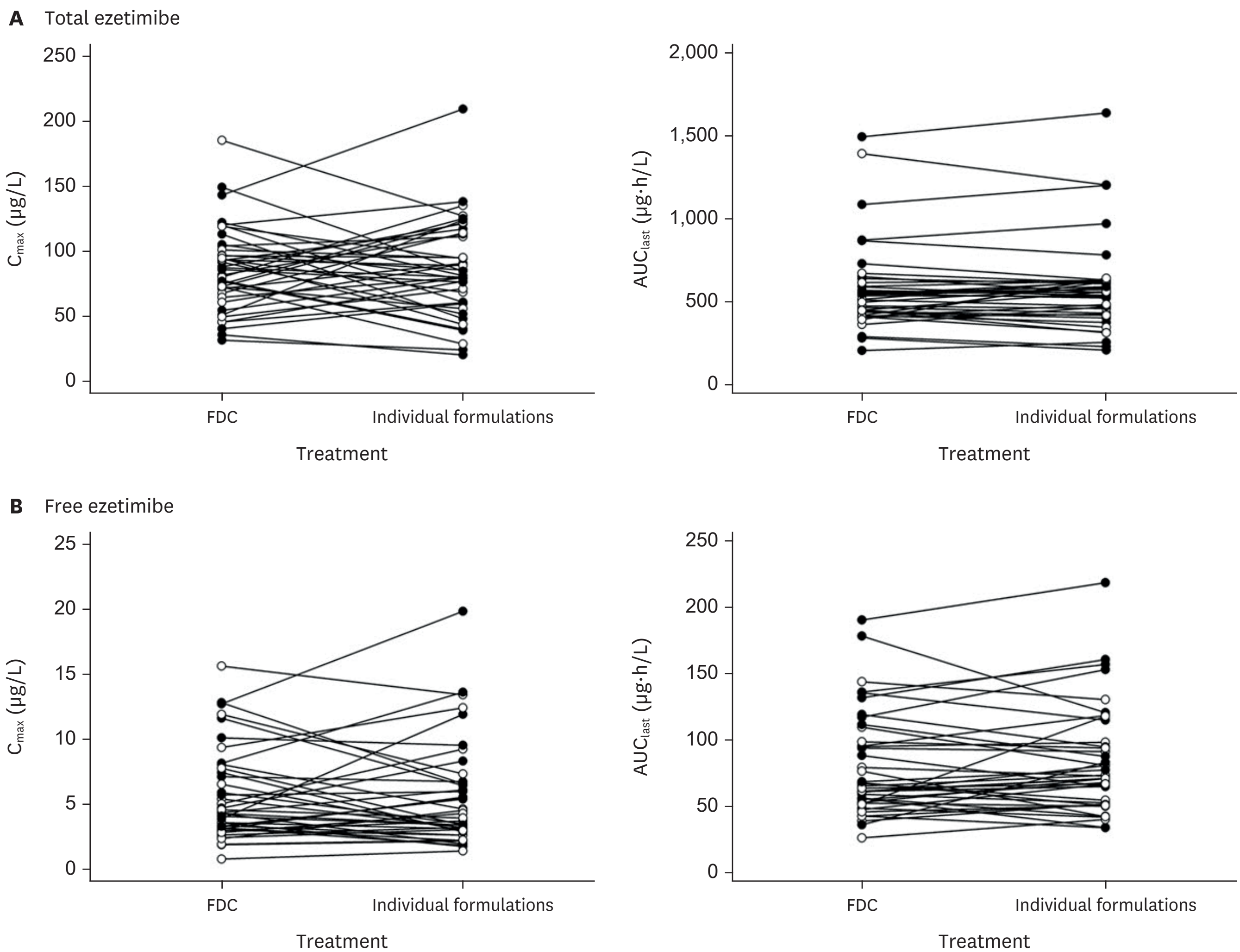


**Figure 4.** Individual comparison for $C_{max}$ and $AUC_{last}$ of (A) total ezetimibe and (B) free ezetimibe after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations.
$C_{max}$, maximum plasma concentration; $AUC_{last}$, area under the curve from zero to the last measurable time point; FDC, fixed-dose combination.

# DISCUSSION

This study compared the PK characteristics of the FDC of rosuvastatin/ezetimibe 2.5/10 mg and the co-administration of individual formulations. In healthy Korean subjects, the FDC showed similarity of PK profiles to the co-administration of each formulation, and the GMR and its 90% CI values for the PK parameters of rosuvastatin and ezetimibe were all within the range of bioequivalence criteria. These results confirm the PK equivalence between the FDC and the co-administration of individual formulations.

Following oral administration in humans, ezetimibe is rapidly absorbed by the intestine and extensively metabolized (> 80%) to its pharmacologically active glucuronide conjugate [12]. According to bioequivalence study guidance from FDA and MFDS, it is recommended to measure only the parent drug rather than metabolites, because the concentration-time profile of the parent drug is more likely to be changed in formulation performance than a metabolite [22]. However, in the case of drugs like ezetimibe that show rapid pre-systemic metabolism, the bioequivalence guidance from the regulatory agencies recommends measuring both parent drug and its primary metabolite(s). In addition, the ezetimibe-specific guidance for generic drug development from FDA and ezetimibe-specific bioequivalence guidance from MFDS suggest that unconjugated ezetimibe and/or total ezetimibe (unconjugated ezetimibe + ezetimibe glucuronide) in plasma can be measured [23,24]. Therefore, this study measured not only free (unconjugated) ezetimibe but also total ezetimibe, in accordance with the regulatory recommendations. Both free (unconjugated) and total ezetimibe were confirmed to be equivalent between the FDC and the co-administration of individual formulations.

In this study, 50 subjects were initially estimated as the target sample size, considering a dropout rate of 20%; however, only 41 subjects completed the study. Nonetheless, based on previous literature, 40 subjects were considered adequate for the purpose of this study, indicating that 41 subjects were sufficient. In addition, the intra-subject CV obtained from this study was 22.2% and 18.3% for the $C_{max}$ and $AUC_{last}$ of rosuvastatin, respectively, while the corresponding values for total ezetimibe were 33.8% and 12.7%, respectively. Considering the highest CV (33.8%) and observed GMR in this study which was close to 1 (**Tables 1** and **2**), a sample size of 37 subjects was determined using the FDC to co-administration ratio of 1.0, with 80% statistical power at a 5% significance level to show whether the PK parameters were different more than 20% between treatments. Overall, these findings based on both the literature and the intra-subject variability observed in this study supported that the 41 subjects were sufficient to assess equivalence between the FDC and the co-administration.

One subject showed notably higher systemic exposure to rosuvastatin compared to overall subjects, with $C_{max}$ and $AUC_{last}$ values approximately 4.5- and 2.9-fold higher, respectively, than the corresponding mean values. Similarly, the exposures of both total and free ezetimibe were also elevated in this subject relative to the mean value. Despite this increased exposure, the subject did not exhibit any AEs or clinically significant abnormalities following treatment. These findings suggest that the observed increase in systemic exposure did not have a significant impact on the safety and tolerability profile.

In accordance with WHO guidance, the FDC of rosuvastatin/ezetimibe 2.5/10 mg is classified under Scenario 3 for FDC-finished pharmaceutical products [25]. In Scenario 3, bioequivalence data may be required for marketing authorization depending on specific circumstances, while clinical safety and efficacy studies are considered mandatory [25]. Likewise, the MFDS guideline for clinical trials of combination drugs mandates the submission of data on bioequivalence, drug-drug interaction (DDI), and clinical safety and efficacy for marketing approval [26]. In alignment with these regulatory expectations, this study successfully demonstrated the bioequivalence between the FDC and the co-administration of individual formulations. Furthermore, a phase 3 study in patients with primary hypercholesterolemia showed the appropriate safety and efficacy of rosuvastatin/ezetimibe 2.5/10 mg over rosuvastatin 2.5 mg or ezetimibe 10 mg [27]. Although a DDI study is typically required under MFDS guidance, the requirement was waive in this case, as the DDI potential had already been evaluated and confirmed through the previously approved FDCs of rosuvastatin/ezetimibe in the doses of 5/10, 10/10, and 20/10 mg [27]. As a result, the FDC of rosuvastatin/ezetimibe 2.5/10 mg has been approved and is now commercially available as LOWROZE Tab. 10/2.5 mg (Addpharma, Seongnam, Korea) or ROSUVAMIBE Tab. 10/2.5 mg (Yuhan, Seoul, Korea), providing an additional option for managing hypercholesterolemia in Korea.

In conclusion, the PK of the FDC of rosuvastatin/ezetimibe 2.5/10 mg was equivalent to the co-administration of individual formulations and was well-tolerated in healthy Korean subjects.

# SUPPLEMENTARY MATERIAL

## Supplementary Table 1

Summary of the treatment-emergent adverse events after a single oral administration of the FDC of rosuvastatin/ezetimibe 2.5 mg/10 mg or the co-administration of individual formulations

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